Alexia Corradini

email: ac2478@cam.ac.uk – webpage

I am a final year PhD Student at the University of Cambridge, working in the field of symplectic topology with Prof. Ivan Smith. My research involves using tools from algebraic geometry and mirror symmetry to understand Lagrangians in symplectic manifolds. Very broadly, I am passionate about geometry and mathematical physics.

Education

2023-current PhD student in Mathematics at the University of Cambridge, supervised by Prof. Ivan Smith
2023 M2 thesis on Seiberg-Witten theory supervised by Prof. Joel Fine at *Université Libre de Bruxelles*
2022-2023 M2 in Analysis, Arithmetic and Geometry at *Université Paris-Saclay - honours*
2022 Research student at the University of Cambridge, supervised by Dr. Ruadhaí Dervan (Spring/Summer)
2021-2022 M1 Mathematics (Jacques Hadamard programme) at *Ecole Polytechnique*
2018-2021 BSc in Mathematics at *Université Libre de Bruxelles - high honours*
2018-2021 BSc in Physics at *Université Libre de Bruxelles - high honours*

Papers & preprints

2026 Algebraically trivial tropical cycles are smash-nilpotent. *Preprint.*

2025 Algebraic Lagrangian cobordisms, flux and the Lagrangian Ceresa cycle. Published in *Journal für die reine und angewandte Mathematik (Crelle's Journal).*

2022 Equivariant localisation in the theory of Z-stability for Kähler manifolds. Published in *International Journal of Mathematics*.

Awards & Prizes

2026 Rayleigh-Knight Prize - *University of Cambridge*
2023 Junior Maryam Mirzakhani prize - *Fondation Mathématique Jacques Hadamard*
2022 Research Internship Prize - *Ecole Polytechnique*

Talks

15/12/2026 "Smash-nilpotence: algebraic,tropical and symplectic" - Symplectic geometry seminar, University of Bonn (*Upcoming*).
04/12/2026 "Smash-nilpotence for Lagrangian submanifolds" - Symplectix Paris (*Upcoming*).
02/12/2026 "Smash-nilpotence: algebraic,tropical and symplectic" - Algebraic geometry seminar, University of Warwick (*Upcoming*).
09/11/2026 "Smash-nilpotence: algebraic,tropical and symplectic" - Symplectic geometry seminar, ETH Zürich.
03/11/2026 "Smash-nilpotence: algebraic,tropical and symplectic" - Algebraic and Symplectic geometry seminar, University of Oxford.
18/06/2026 "A Lagrangian cycle theory" - Algebraic geometry seminar, Institut de Mathématiques de Jussieu
25/03/2026 "A Lagrangian cycle theory" - Conference: Floer theory beyond Floer, University of Cambridge
13/05/2025 "Towards a Lagrangian cycle theory: the Lagrangian Ceresa cycle" - Algebraic geometry seminar, University of Cambridge

15/10/2025 "The Lagrangian Ceresa cycle" - EDGE seminar, University of Edinburgh

13/05/2025 "The Lagrangian Ceresa cycle" - Algebraic geometry seminar, University of Glasgow

28/04/2025 "The Lagrangian Ceresa cycle" - MAGIC seminar, Imperial College London

24/04/2025 "Algebraic Lagrangian cobordisms" - Syzygies and mirror symmetry virtual seminar

11/04/2025 "The Lagrangian Ceresa cycle" - Symplectic Zoominar

21/03/2025 Expository talk on the paper Generating the Fukaya category for Hamiltonian G-manifolds by Evans–Lekili at the European Kylerec workshop in Uppdal, Norway

07/03/2025 "Holomorphic curve theory and the topology of Lagrangians" - Junior geometry seminar, Imperial College London

29/04/2024 Expository talk on *Flow categories* at the Spring School on Floer homotopy theory in Pont-sur-Yonne, France

11/08/2022 "Equivariant localization in the theory of Z-stability and Z-critical metrics" - Cambridge complex geometry afternoon, University of Cambridge

17/06/2022 "Localization theorems in Kähler geometry" - Junior Geometry Seminar, University of Cambridge

Events organised

July 2025 Together with Danil Koževnikov, I organised a summer school on partially wrapped Fukaya categories which took place in Helmsley, UK and was mentored by Sheel Ganatra and Jeff Hicks. We hosted 29 participants - mainly PhD students or early career postdocs.

2024-2025 Together with Adrian Dawid, we organised an internal A-side seminar over three terms, bringing together the symplectic group at the University of Cambridge, offering opportunities for members to present original research, or discuss an interesting paper.

Teaching

2025& 2026 University of Cambridge - Lecturer for Part III preparatory workshops in algebraic topology

2024-2025 University of Cambridge - Supervisor for Part II *Algebraic topology*

2023-2024 University of Cambridge - Supervisor for Part II *Algebraic Topology*

2023-2024 University of Cambridge - Supervisor for Part II *Riemann surfaces*

2022-2023 Ecole Polytechnique - Tutor for first year analysis

Last updated: November 9, 2026